Multi-Epoch Scaling Laws: Three Functional Forms

Setup

We train a 30M-parameter model on DCLM data at 8 chinchilla scales (0.05x – 16x, where 1x = 600M tokens) and up to 128 epochs per scale. All forms share the classic 1-epoch Chinchilla baseline fitted on scales $\geq$ 0.5x:

$$L = E + \frac{B}{D^\beta}, \quad E = 3.72,\ B = 1{,}421{,}053,\ \beta = 0.682,\ R^2 = 0.992$$

Form 1: Ad-Hoc (Token Discount)

$$L = E' + \frac{B}{(D + \eta \cdot D')^\alpha}$$

where $D' = (e-1) \cdot D$ is additional tokens from repetition, and η is a discount factor on those extra tokens.

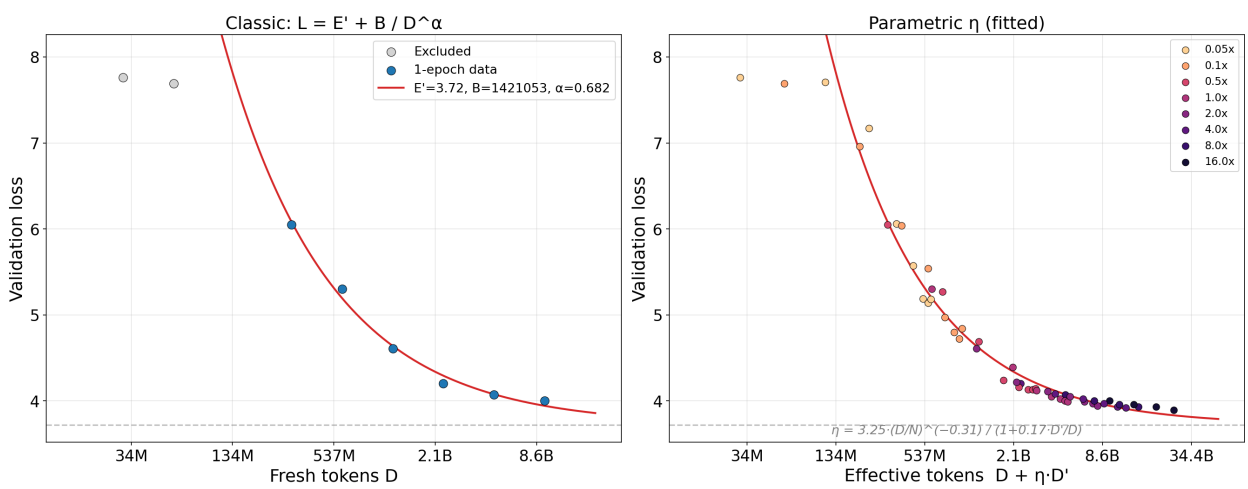
Fitting: E', B, α from 1-epoch data; then fit η as a parametric function:

$$\eta(D, D') = \frac{c \cdot \left(\frac{D}{N}\right)^{-\gamma}}{1 + \beta_{\text{sat}} \cdot \frac{D'}{D}}$$

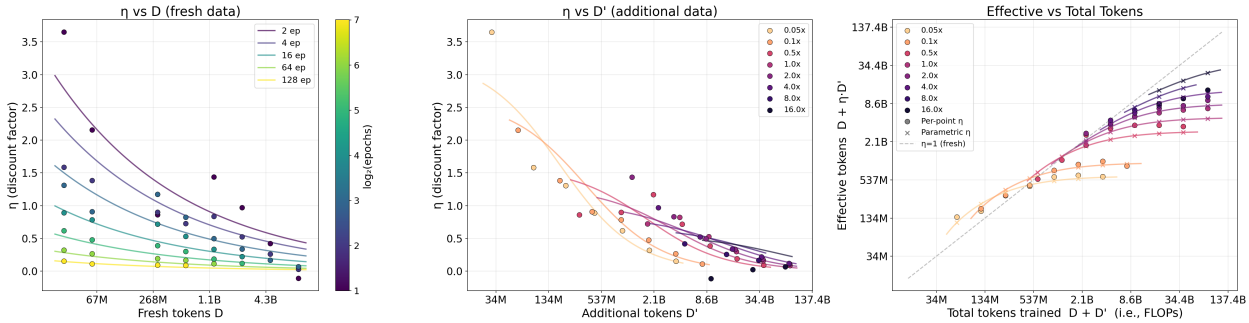
This is Michaelis-Menten in D'/D: $\eta \cdot D'$ saturates as $D' \to \infty$. The D-dependence captures that small datasets benefit more from repetition.

Key features: - Per-point η can be solved analytically from each data point - Parametric η captures both data-richness decay and saturation - 3 params for η on top of the 3 classic params

Results:



Left: 1-epoch fit. Right: parametric η collapse (all points mapped to effective tokens).



Left: η vs D. Middle: η vs D'. Right: effective tokens $D + \eta \cdot D'$ vs total tokens $D + D'$ (i.e., FLOPs). Circles = per-point η, crosses = parametric η, lines = parametric fit.

Extension to paraphrasing: - D' becomes a mix of repeated and paraphrased tokens - Refit constants in η for each pretraining data strategy (repeat, paraphrase, etc.) and directly compare the resulting η curves - Main advantage: provides a common currency for comparing pretraining data strategies - Limitation: conflates the epoch effect and data-quality effect into a single η – hard to disentangle "is paraphrasing better because it's higher quality, or because it reduces effective repetition?"

Form 2: Envelope (Shifted Power Law)

$$L_{\min} = E + \frac{B_{\text{multi}}}{D^{\beta_{\text{multi}}}}$$

For each fresh data size D, find the minimum loss across all epoch counts. Fit a second power law to these envelope points, sharing E from the 1-epoch fit.

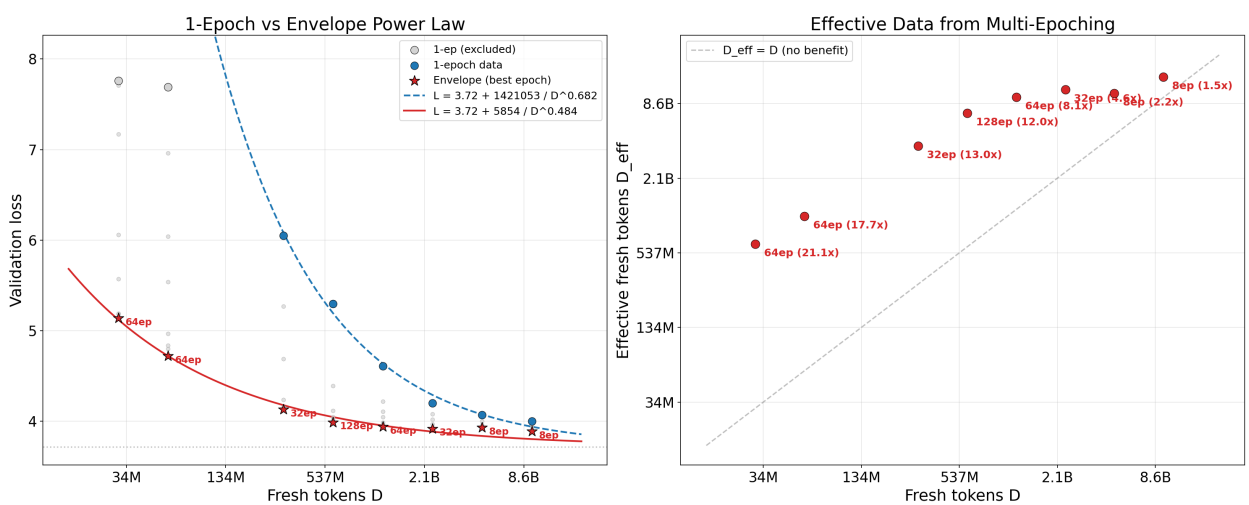
Fitting: E fixed from 1-epoch; fit B_{multi}, β_{multi} on all 8 envelope points.

$$B_{\text{multi}} = 5854, \quad \beta_{\text{multi}} = 0.484, \quad R^2 = 0.985$$

The exponent drops from 0.682 (1-epoch) to 0.484 (envelope), meaning loss decreases more slowly with D when you're already multi-epoching optimally.

Key features: - Simplest form: just two power laws - Completely abstracts away epoch count (compute)

Results:



Left: both power laws. Stars = envelope points annotated with optimal epoch count. Right: D_{eff} vs D showing the value of multi-epoching.

Extension to paraphrasing: The form becomes

$$L_{\text{min,para}} = E + \frac{B_{\text{multi}}}{(D + \eta_{\text{para}} \cdot D')^{\beta_{\text{multi}}}}$$

- Requires running many (D, D') combinations through many epochs to find $L_{\min}$ per combo, then fitting η_{para}
- Unclear whether β_{multi} from repeat-only data transfers to the mixed setting
- Two-stage fitting (first envelope, then η_{para}) propagates errors – envelope R^2=0.985 means residuals distort downstream η_{para}
- In practice: per-point η_{para} ranged from 0.11 to 0.91 across 4 scales, with 1.0x as a clear outlier
- Operationally annoying: need to sweep epochs at every (D, D') combo just to find the envelope

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Form 3: Epoch Discount

$$L = E + \frac{B}{(\eta_{\text{epoch}}(e, D) \cdot D)^{\beta}}$$

where η_{epoch} is an effective epoch multiplier:

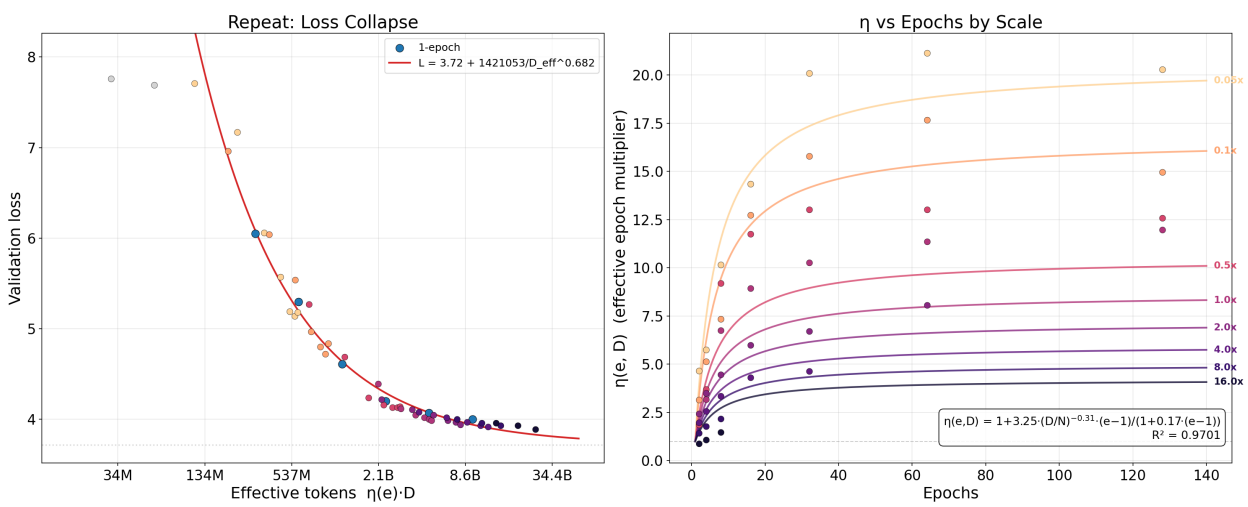
$$\eta_{\text{epoch}}(e, D) = 1 + a \cdot \left(\frac{D}{N}\right)^{-\gamma} \cdot \frac{e - 1}{1 + b \cdot (e - 1)}$$

Fitting: E, B, β from 1-epoch; then fit a, γ, b on all 44 multi-epoch data points.

$$a = 3.25, \quad \gamma = 0.31, \quad b = 0.17, \quad R^2 = 0.970$$

Key features: - Models each (scale, epoch) pair, not just the envelope - $\eta(1, D) = 1$ exactly (baseline) - Michaelis-Menten saturation: $\eta(\infty, D) = 1 + \frac{a}{b} \cdot \left(\frac{D}{N}\right)^{-\gamma}$ - Saturation ceiling depends on D: ~20x at 0.05x scale, ~4x at 16x scale

Results:



Left: loss collapse after mapping to effective tokens. Right: η vs epochs by scale with fitted curves.

Extension to paraphrasing: The natural form is $L = E + \frac{B}{(\eta_{\text{epoch}}(e,\ D+D')\cdot(D+\eta_{\text{para}}\cdot D'))^{\beta}}$ where $e = \frac{\text{epochs}\cdot D}{D+D'}$ is effective passes through the mixed pool.

- Cleanest separation of epoch effect (η_{epoch}) from data-quality effect (η_{para})
- Entangled in practice: need enough variation in both epoch count and D'/D ratio to pin them down independently
- With current data (4 scales, fixed $D'/D \approx 0.48$), η_{para} ranges from 1.3 to 4.4 depending on whether η_{epoch} is fixed from repeat or jointly refit
- Unclear whether η_{epoch} from repeat-only training transfers to the mixed setting
- Error propagation from two-stage fit remains a concern

Comparison

	Ad-Hoc	Envelope	Epoch Discount
What it models	Discount on extra tokens D'	Best achievable loss per D	Effective epoch multiplier
Granularity	Per (D, D') pair	Per D (envelope only)	Per (D, e) pair
Free params (beyond 1-ep)	3 $(c, \gamma, \beta_{\text{sat}})$	2 $(B_{\text{multi}}, \beta_{\text{multi}})$	3 (a, γ, b)
R^2	~0.97 (on loss)	0.985 (on envelope)	0.970 (on all multi-epoch)
Abstracts away epochs?	Partially (via D')	Completely	No (explicit input)
Para extension	Refit η per strategy; direct comparison	Needs many (D, D') combos; error propagates	Needs varied D'/D ratios; identifiability issue

All three tell a consistent story: multi-epoching provides substantial benefit at small D (up to ~20x effective data) with diminishing returns, and the benefit shrinks at larger D.

What experiments would help most: - More paraphrase scales (especially larger D where repeat vs para differences are more pronounced) - Varying the D'/D ratio (e.g., $D' = 0.25D,\ 0.5D,\ 1.0D,\ 2.0D$) at fixed D to disentangle η_{epoch} from η_{para} - 370M model results to check if the fitted constants transfer across model sizes